European option pricing: The Black-Scholes PDE

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Contents

1 Introduction **1**

2 Setup: definitions and objectives **2**
2.1 Definitions and financial context . . . 2
2.2 Project goals . . . 2

3 The Black-Scholes model **2**
3.1 Black-Scholes PDE . . . 4
3.2 Transformation to log-price . . . 4
3.3 Transformation to the heat equation . . . 5

4 Existence and uniqueness of a solution **6**

5 Numerical experiments (FDM) **7**
5.1 Finite differences method . . . 7
5.1.1 Space semi–discretization . . . 7
5.1.2 Explicit Euler . . . 9
5.1.3 Implicit Euler . . . 13
5.1.4 Crank-Nicholson . . . 15
5.2 Greeks . . . 16

6 Binomial tree pricing (Cox-Ross-Rubinstein model) **17**
6.1 Model framework . . . 17
6.2 Implementation . . . 18
6.3 Numerical experiment . . . 18

1 Introduction

In this project, we focus on the valuation of European options, in particular the European call option, which gives its holder the right to purchase an asset at a predetermined strike price at a fixed maturity date.

While the Black–Scholes model admits an analytical solution, many financial derivatives require numerical approximation techniques. In this work, we study the theoretical well-posedness of the governing partial differential equation (PDE) and compare several numerical schemes, including finite difference methods (FDM) with different Euler discretizations and binomial tree methods.

2 Setup: definitions and objectives

This section is inspired by material from CIP's course and N. Hilber et al., *Computational Methods for Quantitative Finance*.

2.1 Definitions and financial context

A European call option is a financial contract giving the holder the right, but not the obligation, to buy an underlying asset S at a strike price K at maturity time T. We introduce the following quantities:

- S_t: price of the underlying asset at time t,
- K: strike price,
- r: constant risk-free interest rate,
- σ: volatility of the underlying asset,
- $g(S) = \max(S - K, 0)$: payoff function at maturity.

2.2 Project goals

The objectives of this companion project are:

1. Setting up the problem : Derive the PDE from the underlying stochastic differential equation using Itô calculus and the infinitesimal generator (from CIP's course and N. Hilber et al., *Computational Methods for Quantitative Finance*).
2. Theoretical analysis: Establish the well-posedness (existence, uniqueness and stability in the sense of Hadamard) of the Black-Scholes PDE in appropriate functional spaces.
3. Numerical implementation:
    - Price a European call option using publicly available market data for NVIDIA.
    - Solve the PDE using finite difference methods (explicit, implicit and Crank-Nicolson schemes).
    - Study the influence of boundary conditions and discretization on the stability and accuracy of the numerical solvers.
    - Compare results with the binomial tree model.
    - Compute some greeks associated with the FDM

3 The Black-Scholes model

We derive the Black-Scholes partial differential equation for a European call option written on a non-dividend-paying stock with strike price K and maturity T.

Model assumptions The Black-Scholes model relies on the following assumptions:

- The stock price follows a geometric Brownian motion with constant drift and volatility.
- There are no transaction costs or taxes.
- The stock pays no dividends during the life of the option.

- Markets are frictionless and arbitrage-free.
- The risk-free interest rate r is constant and known.
- Options are European-style and can only be exercised at maturity.
- Trading is continuous in time and can be done in arbitrary quantities.

Stock price Let $(S_t)_{t\geq 0}$ denote the price of the stock at time t. We assume that it satisfies the stochastic differential equation

$$dS_t = \mu S_t\, dt + \sigma S_t\, dW_t, \tag{1}$$

where $(W_t)_{t\geq 0}$ is a standard Brownian motion, μ is the drift parameter, and $\sigma > 0$ is the volatility.

We also introduce a risk-free cash account $(B_t)_{t\geq 0}$ defined by

$$B_t = e^{rt}, \tag{2}$$

where r is the constant risk-free interest rate.

Let $C(S,t)$ denote the price of a European call option at time t when the stock price is S.

Payoff function The payoff of a European call option at maturity is described by the function

$$g(S) = \max(S-K, 0). \tag{3}$$

Thus, the terminal condition is

$$C(S,T) = g(S). \tag{4}$$

Itô's Lemma

Theorem 3.1 (Itô's Lemma, admitted)**.** *Let $(X_t)_{t\geq 0}$ satisfy*

$$dX_t = \mu X_t\, dt + \sigma X_t\, dW_t,$$

and let $f \in C^{2,1}(\mathbb{R}\times[0,T])$. Then the process $Y_t = f(X_t,t)$ satisfies

$$\begin{aligned} df(X_t,t) = &\left(\frac{\partial f}{\partial t} + \mu X_t \frac{\partial f}{\partial x} + \frac{1}{2}\sigma^2 X_t \frac{\partial^2 f}{\partial x^2}\right) dt \\ &+ \sigma X_t \frac{\partial f}{\partial x} dW_t. \end{aligned} \tag{5}$$

Applying Itô's lemma to $C(S_t,t)$ gives

$$dC(S_t,t) = \Big(\frac{\partial C}{\partial t} + \mu S_t \frac{\partial C}{\partial S} + \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 C}{\partial S^2}\Big) dt + \sigma S_t \frac{\partial C}{\partial S} dW_t. \tag{6}$$

Self-financing portfolio We construct a self-financing trading strategy consisting of:

- x_t units of the cash account,
- y_t units of the stock.

The value of the portfolio at time t is

$$P_t = x_t B_t + y_t S_t. \tag{7}$$

The self-financing condition implies

$$dP_t = x_t\, dB_t + y_t\, dS_t. \tag{8}$$

Using the dynamics of B_t and S_t, we obtain

$$\begin{aligned} dP_t &= r x_t B_t\, dt + y_t(\mu S_t dt + \sigma S_t dW_t) \\ &= (r x_t B_t + \mu y_t S_t) dt + \sigma y_t S_t dW_t. \end{aligned} \tag{9}$$

3.1 Black-Scholes PDE

We choose the strategy (x_t, y_t) such that the portfolio replicates the option price:

$$P_t = C(S_t, t).$$

Equating the diffusion and drift terms of $dC(S_t, t)$ and dP_t yields

$$y_t = \frac{\partial C}{\partial S}, \tag{10}$$

$$r x_t B_t = \frac{\partial C}{\partial t} + \frac{1}{2}\sigma^2 S_t^2 \frac{\partial^2 C}{\partial S^2}. \tag{11}$$

Substituting into $P_t = x_t B_t + y_t S_t = C(S_t, t)$ gives the Black–Scholes PDE:

$$\frac{\partial C}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 C}{\partial S^2} + rS\frac{\partial C}{\partial S} - rC = 0, \quad (t, S) \in [0, T) \times \mathbb{R}_+. \tag{12}$$

Boundary conditions For a European call option, the PDE is supplemented with the boundary condition coming from the exercise of the option:

$$C(S, T) = g(S) = \max(S - K, 0), \tag{13}$$

$$C(0, t) = 0 \tag{14}$$

3.2 Transformation to log-price

In order to facilitate both the theoretical analysis and the numerical approximation of the Black–Scholes equation, it is convenient to introduce a change of variables based on the logarithm of the stock price. The log-price transformation converts the Black-Scholes equation into a linear parabolic equation with constant coefficients. Then we use an appropriate change of variables to transform the Black-scholes equation into the familiar heat equation studied in class.

We define the log-price variable and the time-to-maturity by

$$x = \log(S), \qquad \tau = T - t,$$

and we set

$$v(\tau, x) = C(t, S).$$

Transformed equation. Using the chain rule, the Black-Scholes PDE becomes

$$v_\tau - \frac{\sigma^2}{2} v_{xx} - \left(r - \frac{\sigma^2}{2}\right) v_x + rv = 0 \quad \text{in } (0, T] \times \mathbb{R}, \tag{15}$$

The initial condition is obtained from the payoff function:

$$v(0, x) = g(e^x) = \max(e^x - K, 0). \tag{16}$$

Spatial domain. After the logarithmic transformation $x = \ln S$, the spatial domain becomes $\mathbb{R}$. For numerical purposes, this unbounded domain is truncated to abounded interval.

Let

$$G := (x_{\min}, x_{\max}) \subset \mathbb{R},$$

where $x_{\min} < x_{\max}$ are fixed constants. This corresponds in the original asset variable to the artificial bounds

$$S_{\min} = e^{x_{\min}}, \qquad S_{\max} = e^{x_{\max}}.$$

In practice, we'll pick $x_{\min} << \ln(K)$, and $x_{\max} >> \ln(K)$.

3.3 Transformation to the heat equation

To transform the equation into a heat equation, we introduce the change of variables

$$v(\tau, x) = e^{\alpha x + \beta \tau} u(\tau, x),$$

where the constants α and β will be chosen to eliminate the u_x and u terms.

Step 1: compute derivatives. Using the product rule, we obtain

$$v_\tau = e^{\alpha x + \beta \tau}(u_\tau + \beta u),$$

$$v_x = e^{\alpha x + \beta \tau}(u_x + \alpha u),$$

$$v_{xx} = e^{\alpha x + \beta \tau}(u_{xx} + 2\alpha u_x + \alpha^2 u).$$

Step 2: substitute into the PDE Substituting these expressions into the transformed Black–Scholes equation yields

$$e^{\alpha x + \beta \tau}(u_\tau + \beta u) - \frac{\sigma^2}{2} e^{\alpha x + \beta \tau}(u_{xx} + 2\alpha u_x + \alpha^2 u) \tag{17}$$

$$- \left(r - \frac{\sigma^2}{2}\right) e^{\alpha x + \beta \tau}(u_x + \alpha u) + r e^{\alpha x + \beta \tau} u = 0. \tag{18}$$

Dividing by $e^{\alpha x + \beta \tau}$ gives

$$u_\tau + \beta u - \frac{\sigma^2}{2} u_{xx} - \sigma^2 \alpha u_x - \frac{\sigma^2}{2}\alpha^2 u - \left(r - \frac{\sigma^2}{2}\right) u_x - \left(r - \frac{\sigma^2}{2}\right) \alpha u + r u = 0. \tag{19}$$

Step 3: organize Grouping the terms according to u_{xx}, u_x, and u gives

$$u_\tau - \frac{\sigma^2}{2} u_{xx} - \left(\sigma^2 \alpha + r - \frac{\sigma^2}{2}\right) u_x + \left(\beta - \frac{\sigma^2}{2}\alpha^2 - \alpha\left(r - \frac{\sigma^2}{2}\right) + r\right) u = 0. \tag{20}$$

Step 4: chose the parameters To eliminate the u_x term we choose α such that

$$\sigma^2 \alpha + r - \frac{\sigma^2}{2} = 0,$$

which yields

$$\alpha = -\frac{r - \sigma^2/2}{\sigma^2}.$$

To eliminate the u term we require

$$\beta - \frac{\sigma^2}{2}\alpha^2 - \alpha\left(r - \frac{\sigma^2}{2}\right) + r = 0.$$

Substituting the value of α gives

$$\beta = -\frac{(r - \sigma^2/2)^2}{2\sigma^2} - r.$$

Step 5: the heat equation With these choices of α and β, the equation reduces to

$$u_\tau - \frac{\sigma^2}{2} u_{xx} = 0, \tag{21}$$

which is the heat equation.

Step 6: Initial conditions Using the transformation

$$v(\tau, x) = e^{\alpha x + \beta \tau} u(\tau, x),$$

the initial condition at $\tau = 0$ reads

$$u(0, x) = e^{-\alpha x} v(0, x).$$

From the payoff function $g(S) = \max(S - K, 0)$, we obtain

$$u(0, x) = e^{-\alpha x} g(e^x) = e^{-\alpha x} \max(e^x - K, 0).$$

Final formulation Let $\Omega = G = (x_{\min}, x_{\max}) \subset \mathbb{R}$ denote the truncated spatial domain. Define the initial data

$$u_0(x) := e^{-\alpha x} \max(e^x - K, 0) \in L^2(\Omega).$$

The transformed PDE can then be written in the standard heat equation form

$$\begin{cases} \dfrac{du}{d\tau} - \dfrac{\sigma^2}{2} u_{xx} = 0, & (\tau, x) \in (0, T) \times \Omega, \\ u(\tau, x_{\min}) = 0, \quad u(\tau, x_{\max}) = 0, & \tau \in (0, T), \\ u(0, .) = u_0 \in L^2(\Omega). \end{cases}$$

4 Existence and uniqueness of a solution

Existence and uniqueness. Since Ω is bounded and open subset of $\mathbb{R}$, and $u_0 \in L^2(\Omega)$, standard results for parabolic PDEs (from the class) guarantee that there exists a unique solution

$$u \in L^2(0, T; H_0^1(\Omega)) \cap C^0([0, T]; L^2(\Omega))$$

Here $H_0^1(\Omega)$ is the standard Sobolev space with zero trace at the boundary.
More, the solution satisfies:

$$\forall t, \|u(t)\|^2_{L^2(\Omega)} + \int_0^t \|u(s)\|^2_{H_0^1(\Omega)} \, ds \leq C \|u_0\|_{L^2(\Omega)}$$

for some constant $C > 0$ independent of u_0.

Note that, because $f = 0$, we have a regularizing effect: the solution u is smoother than the data $u0$:

$$u \in C^\infty\big(\mathbb{R}_+^* \times \overline{\Omega}\big)$$

There is a closed-form solution, that I won't prove here but use to compare my numerical approximations.

Solution The unique solution of the Black–Scholes PDE for a European call option will be admitted and is

$$C(S,t) = S\Phi(d_1) - Ke^{-r(T-t)}\Phi(d_2), \tag{22}$$

where

$$d_1 = \frac{\ln(S/K) + (r + \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}}, \tag{23}$$

$$d_2 = d_1 - \sigma\sqrt{T-t}, \tag{24}$$

and $\Phi(\cdot)$ denotes the cumulative distribution function of the standard normal distribution.

5 Numerical experiments (FDM)

In this section, we'll introduce the semi-discretization using the FDM, the FEM, and binomial trees.

5.1 Finite differences method

We consider the heat equation obtained after the Black–Scholes transformation:

$$u_\tau - \frac{\sigma^2}{2}u_{xx} = 0 \qquad (\tau, x) \in (0,T) \times \Omega$$

with homogeneous Dirichlet boundary conditions

$$u(\tau, x_{\min}) = 0, \qquad u(\tau, x_{\max}) = 0$$

and initial condition

$$u(0,x) = u_0(x).$$

Let us denote

$$\alpha = \frac{\sigma^2}{2}.$$

The PDE can be rewritten as

$$u_\tau = \alpha u_{xx}.$$

5.1.1 Space semi–discretization

Let

$$\Omega = (x_{\min}, x_{\max}) \subset \mathbb{R}.$$

We introduce a uniform spatial grid.
Let $J \in \mathbb{N}^*$ be the number of interior nodes and define

$$h_x = \frac{x_{\max} - x_{\min}}{J+1}.$$

The grid points are

$$x_j = x_{\min} + jh_x, \qquad j = 0, \ldots, J+1.$$

The boundary nodes are

$$x_0 = x_{\min}, \qquad x_{J+1} = x_{\max}.$$

Because of the Dirichlet boundary conditions we have

$$u(\tau, x_0) = u(\tau, x_{J+1}) = 0.$$

Finite difference approximation For interior nodes x_j with $j = 1, \ldots, J$, we approximate the second derivative using the centered finite difference:

$$u_{xx}(\tau, x_j) \approx \frac{u(\tau, x_{j+1}) - 2u(\tau, x_j) + u(\tau, x_{j-1})}{h_x^2}.$$

Substituting into the PDE gives

$$\frac{\partial u}{\partial \tau}(\tau, x_j) = \alpha \frac{u(\tau, x_{j+1}) - 2u(\tau, x_j) + u(\tau, x_{j-1})}{h_x^2}.$$

Define the vector of interior unknowns

$$V(\tau) = \begin{pmatrix} u(\tau, x_1) \\ \vdots \\ u(\tau, x_J) \end{pmatrix} \in \mathbb{R}^J.$$

The semi-discrete problem can be written as a system of ODEs:

$$V'(\tau) = -\alpha A_{h_x} V(\tau)$$

with initial condition

$$V(0) = U_0 = \begin{pmatrix} u_0(x_1) \\ \vdots \\ u_0(x_J) \end{pmatrix}.$$

The matrix The matrix $A_{h_x} \in M_J(\mathbb{R})$ is the matrix seen in class:

$$A_{h_x} = \frac{1}{h_x^2} \begin{pmatrix} 2 & -1 & 0 & \ldots & 0 \\ -1 & 2 & -1 & \ldots & 0 \\ 0 & -1 & 2 & \ldots & 0 \\ \vdots & \vdots & \vdots & \ddots & -1 \\ 0 & 0 & 0 & -1 & 2 \end{pmatrix}.$$

This matrix is symmetric, positive definite, sparse tridiagonal.

Spectrum of the matrix The eigenvalues of A_{h_x} are known explicitly:

$$\lambda_k = \frac{4}{h_x^2} \sin^2\left(\frac{k\pi}{2(J+1)}\right), \qquad k = 1, \ldots, J.$$

Therefore the spectral radius is

$$\rho(A_{h_x}) = \max_k \lambda_k = \frac{4}{h_x^2} \sin^2\left(\frac{J\pi}{2(J+1)}\right) < \frac{4}{h_x^2}.$$

For large J,

$$\rho(A_{h_x}) \approx \frac{4}{h_x^2}.$$

We still need a full discretization, and for that we use and compare Euler implicit, explicit and Crank-Nicholson.

5.1.2 Explicit Euler

We discretize the time interval:

$$\tau_n = nh_t, \qquad n = 0, \dots, N,$$

where

$$h_t = \frac{T}{N}.$$

Let

$$V^n \approx V(\tau_n).$$

Applying the explicit Euler method to

$$V'(\tau) = -\alpha A_{h_x} V(\tau)$$

gives

$$V^{n+1} = V^n - h_t \alpha A_{h_x} V^n.$$

Equivalently,

$$V^{n+1} = (I_J - h_t \alpha A_{h_x}) V^n.$$

Iteration matrix Define

$$B = I_J - h_t \alpha A_{h_x}.$$

Then the scheme becomes

$$V^{n+1} = BV^n.$$

Using the structure of A_{h_x}, the matrix B is

$$B = \begin{pmatrix} 1-2\lambda & \lambda & 0 & \dots \\ \lambda & 1-2\lambda & \lambda & \dots \\ 0 & \lambda & 1-2\lambda & \dots \\ \vdots & \ddots & \ddots & \ddots \end{pmatrix}$$

where

$$\lambda = \alpha \frac{h_t}{h_x^2} = \frac{\sigma^2 h_t}{2h_x^2}.$$

Stability For stability we require

$$\rho(B) \leq 1.$$

Since the eigenvalues of B are

$$\mu_k = 1 - h_t \alpha \lambda_k,$$

we require

$$|1 - h_t \alpha \lambda_k| \leq 1$$

for all k.
The worst case corresponds to

$$\lambda_{\max} \approx \frac{4}{h_x^2}.$$

Thus

$$|1 - h_t \alpha \lambda_{\max}| \leq 1.$$

This gives

$$0 \leq h_t \alpha \lambda_{\max} \leq 2.$$

Substituting the approximation yields

$$h_t \alpha \frac{4}{h_x^2} \leq 2.$$

Hence

$$h_t \leq \frac{h_x^2}{2\alpha}.$$

Since

$$\alpha = \frac{\sigma^2}{2},$$

the CFL condition becomes

$$h_t \leq \frac{h_x^2}{\sigma^2}.$$

Consistency The truncation error satisfies

$$\tau_{n,j} = O(h_t) + O(h_x^2).$$

Thus the scheme is first order in time and second order in space.

Convergence Under the CFL condition

$$h_t \leq \frac{h_x^2}{\sigma^2},$$

the explicit scheme is stable.
By the Lax equivalence theorem, consistency and stability imply convergence.
Therefore

$$\max_{0 \leq n \leq N} \|V^n - \Pi_{h_x} u(\tau_n)\|_2 = O(h_t) + O(h_x^2).$$

Here $\Pi_{h_x} u$ denotes the projection of the exact solution on the grid.

Results We plot the results of the explicit experiments

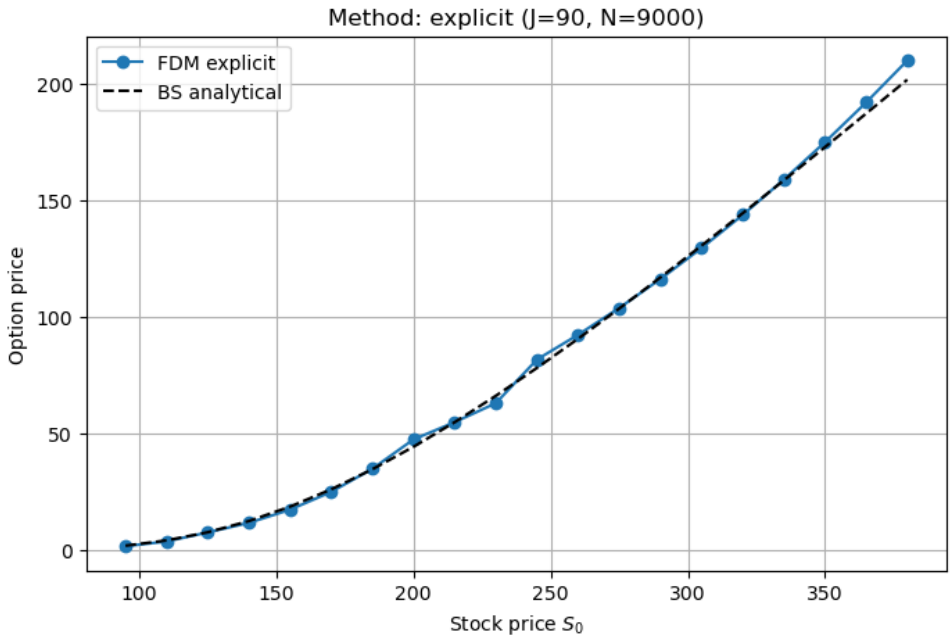

Figure 1: FDM Explicit euler and BS

We can observe that the numerical stock price is very close to the analytical one. This is in the case of stability §CFL condition is verified).

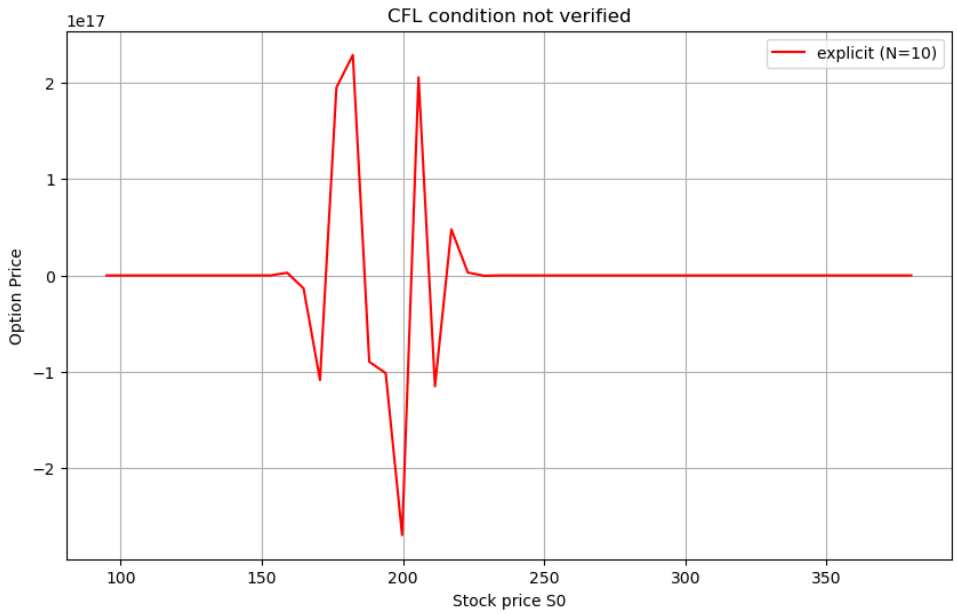

Figure 2: Unstable explicite scheme

We used a very fine spatial grid: $J = 200$, very few time steps: $N = 10$ which violate the CFL condition, creating instability.

The resulting option prices show extremely large oscillations, on the order of

$$\pm 2 \times 10^{17}$$

.

We can also visualize the 3D surface in the case of instability:

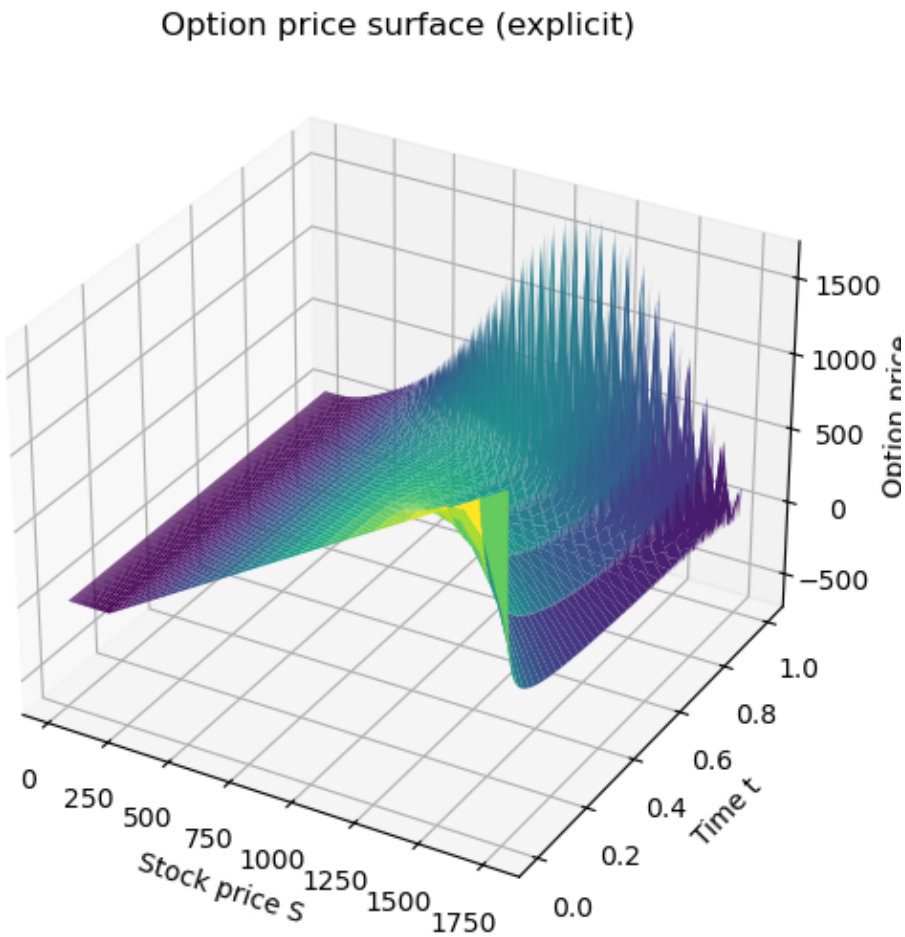

Figure 3: Unstable explicit scheme

5.1.3 Implicit Euler

We discretize the time interval $[0, T]$ into N steps of size $h_t = T/N$, where $\tau_n = nh_t$ for $n = 0, \ldots, N$.

Let $V^n \approx V(\tau_n)$ represent the vector of interior grid points $[v_1^n, \ldots, v_J^n]^T$.
Applying the implicit Euler method to the semi-discrete system $V'(\tau) = -\alpha A_{h_x} V(\tau)$ yields:

$$\frac{V^{n+1} - V^n}{h_t} = -\alpha A_{h_x} V^{n+1} \tag{25}$$

which gives the linear system:

$$(I_J + h_t \alpha A_{h_x}) V^{n+1} = V^n. \tag{26}$$

Stability The method is stable in the $\|\cdot\|_{R_J,2}$ norm. This follows from the eigenvalues of the iteration matrix

$$B = (I_J + h_t\, \alpha A_{h_x})^{-1}, \quad \mu_k = 1/(1 + h_t\, \alpha \lambda_k).$$

Since $\alpha > 0$ and the eigenvalues satisfy $\lambda_k \in (0, 4/h_x^2]$, we have $\rho(B) < 1$ for all $h_t, h_x > 0$. The system matrix

$$L = I_J + h_t\, \alpha A_{h_x}$$

is symmetric positive definite and tridiagonal.

Consistency The local truncation error satisfies

$$E^n = O(h_t) + O(h_x^2),$$

derived via taylor expansions. This confirms that the scheme is first-order accurate in time and second-order accurate in space.

Convergence By the Lax equivalence theorem, the scheme's consistency and stability guarantee convergence of the numerical solution V^n to the discrete projection of the exact solution $\Pi_{h_x} u(\tau_n)$:

$$\max_{0 \le n \le N} \|V^n - \Pi_{h_x} u(\tau_n)\|_2 \le C(h_t + h_x^2), \tag{27}$$

where the constant C depends on the final time T and the regularity of the initial data u_0.

Results We plot the results of the implicit experiments

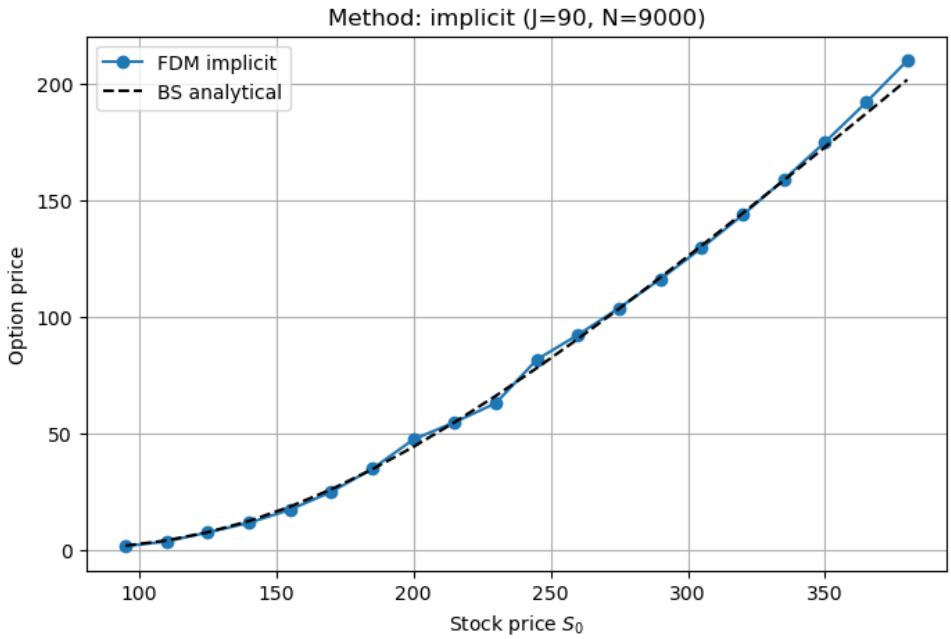

Figure 4: FDM implicit Euler and BS

We can observe that the numerical stock price is very close to the analytical one.

Let's look at the implicit scheme surface. At time $t = 0$, the initial condition presents a spike at the strike price K. The singularity is immediately smoothed out, illustrating the regularizing effect of the PDE, where the solution u becomes C^∞ regardless of the initial data. More, this surface remains smooth and well-behaved, illustrating the stability of the scheme.

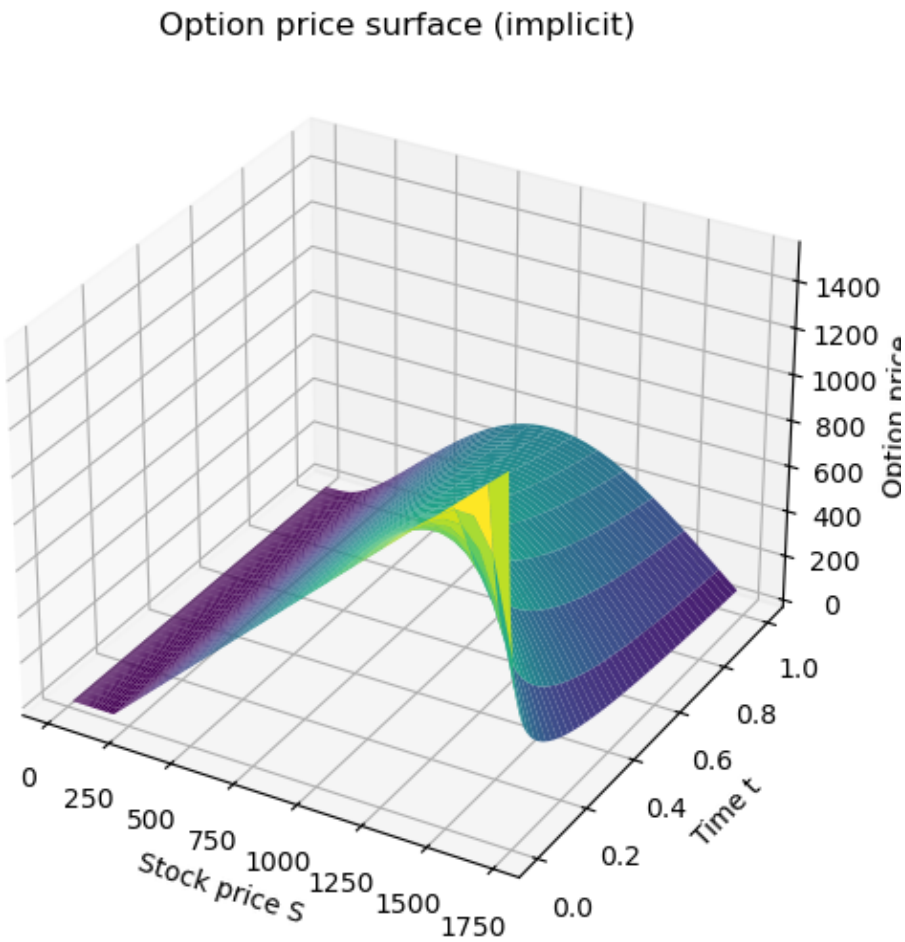

Figure 5: Implicit Euler surface

5.1.4 Crank-Nicholson

The Crank-Nicolson scheme applies a centered difference in time, achieving second-order accuracy. For the spatial discretization, it uses the average at the current and next time steps:

$$\frac{V^{n+1} - V^n}{h_t} = -\alpha \frac{A_{h_x} V^n + A_{h_x} V^{n+1}}{2}. \tag{28}$$

Rewriting as a linear system:

$$\left(I_J + \frac{h_t \alpha}{2} A_{h_x} \right) V^{n+1} = \left(I_J - \frac{h_t \alpha}{2} A_{h_x} \right) V^n. \tag{29}$$

Stability The iteration matrix is

$$B = \left(I_J + \frac{h_t \alpha}{2} A_{h_x} \right)^{-1} \left(I_J - \frac{h_t \alpha}{2} A_{h_x} \right),$$

with eigenvalues

$$\mu_k = \frac{1 - \frac{h_t \alpha}{2} \lambda_k}{1 + \frac{h_t \alpha}{2} \lambda_k}, \quad k = 1, \ldots, J.$$

Since $\lambda_k > 0$, it follows that $|\mu_k| \leq 1$, confirming that the method is stable.

Consistency The local truncation error satisfies

$$E^n = O(h_t^2) + O(h_x^2),$$

showing the scheme is second-order accurate in time and space.

Convergence By the Lax equivalence theorem, stability and consistency imply convergence:

$$\max_{0 \le n \le N} \|V^n - \Pi_{h_x} u(\tau_n)\|_2 = O(h_t^2) + O(h_x^2), \tag{30}$$

where $\Pi_{h_x} u$ denotes the discrete projection of the exact solution.

Results We plot the results of the Crank-Nicholson experiments

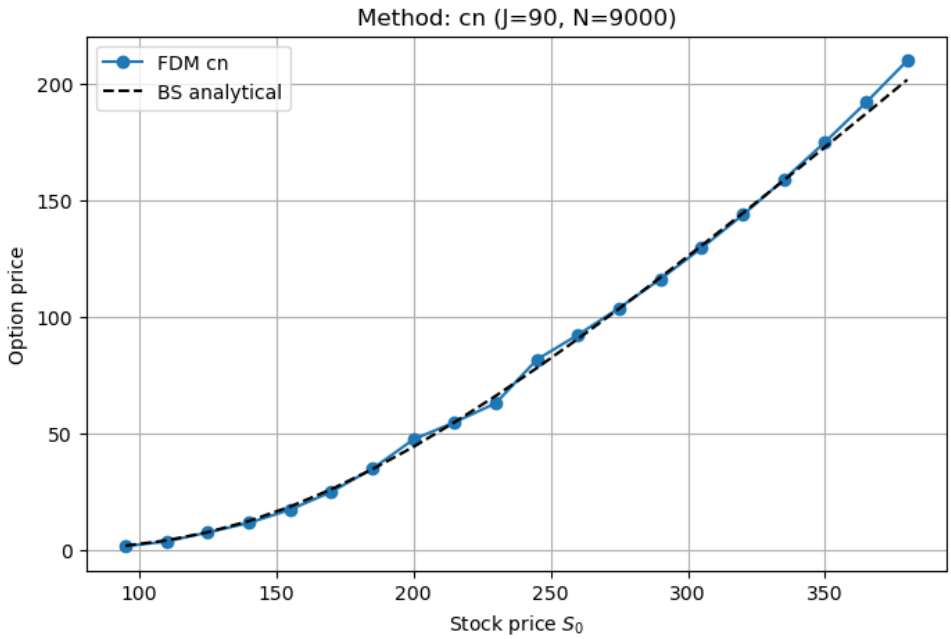

Figure 6: FDM CN Euler and BS

We can observe that the numerical stock price is very close to the analytical one.

5.2 Greeks

In option pricing, the Greeks are sensitivities of the option value with respect to various parameters. They are essential for risk management and hedging strategies.

Once we have computed the option price surface $V(S,t)$ (see the notebook), we can approximate these sensitivities numerically:

Delta (Δ) : measures the sensitivity of the option price to changes in the stock price. It is approximated using central differences along the stock price grid:

$$\Delta(S,t) \approx \frac{V(S+h,t) - V(S-h,t)}{2h}.$$

Gamma (Γ) : represents the rate of change of Delta with respect to the stock price. It is computed via second-order central differences:

$$\Gamma(S,t) \approx \frac{V(S+h,t) - 2V(S,t) + V(S-h,t)}{h^2}.$$

Theta (Θ) : quantifies the sensitivity of the option price to time, using a backward difference in time:

$$\Theta(S,t) \approx \frac{V(S,t) - V(S,t-\Delta t)}{\Delta t}.$$

For an implicit scheme surface, we get the greeks:

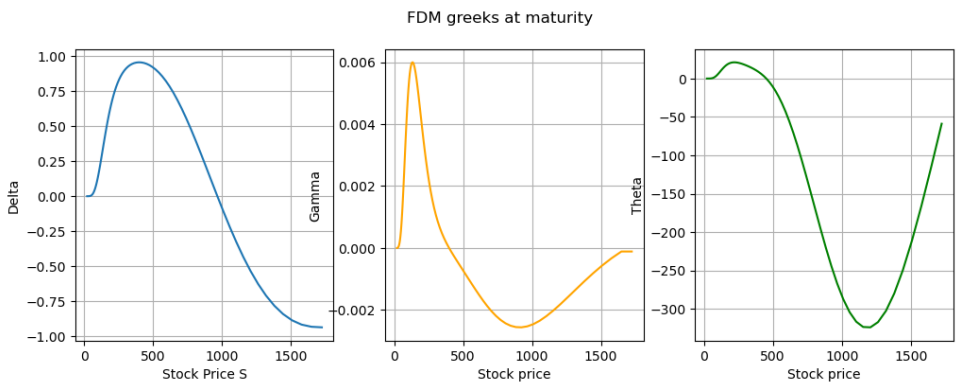

Figure 7: Greeks

6 Binomial tree pricing (Cox-Ross-Rubinstein model)

The binomial tree method is a numerical approach used to approximate the price of a European call option. It models the evolution of the underlying asset price as a lattice, providing an intuitive alternative to Black-Schoels.

6.1 Model framework

The maturity T is divided into N equal time steps $\Delta t = T/N$. Over each step, the asset price S is assumed to move to one of two possible values:

- Up: $S_{up} = S \cdot u$, where $u = e^{\sigma\sqrt{\Delta t}}$.
- Down $S_{down} = S \cdot d$, where $d = 1/u = e^{-\sigma\sqrt{\Delta t}}$.

The probability q is derived from the no-arbitrage principle, which states that under the risk-neutral measure $\mathbb{Q}$, the discounted asset price must be a martingale. This implies that the current price S_0 is the expected value of the future price $S_{\Delta t}$ discounted at the risk-free rate r:

$$S_0 = e^{-r\Delta t}\,\mathbb{E}^{\mathbb{Q}}[S_{\Delta t}]. \tag{31}$$

In the binomial model, the price at the next step can only take two values: $S_0 u$ with probability q, or $S_0 d$ with probability $1-q$. The expectation is:

$$\mathbb{E}^{\mathbb{Q}}[S_{\Delta t}] = q(S_0 u) + (1-q)(S_0 d). \tag{32}$$

Substituting the expectation into the martingale condition:

$$S_0 = e^{-r\Delta t}\left[qS_0 u + (1-q)S_0 d\right]. \tag{33}$$

Dividing by S_0 and multiplying by $e^{r\Delta t}$ gives:

$$e^{r\Delta t} = qu + d - qd \tag{34}$$

$$e^{r\Delta t} - d = q(u - d) \tag{35}$$

$$\implies q = \frac{e^{r\Delta t} - d}{u - d}. \tag{36}$$

Under the risk-neutral measure, the probability of an upward move q is derived to ensure the expected return matches the risk-free rate r:

$$q = \frac{e^{r\Delta t} - d}{u - d}. \tag{37}$$

6.2 Implementation

The pricing process follows a five-step recursive logic:

1. **Time discretization:** The total time to maturity is partitioned into N intervals $\Delta t = T/N$

2. **Terminal asset prices:** At maturity ($t = T$), the asset can reach $N + 1$ possible states. For each $i \in \{0, \ldots, N\}$, the price is:

$$S_{T,i} = S_0 u^{N-i} d^i. \tag{38}$$

3. **Option payoff:** The value of the European call option at each terminal node is given by the payoff function:

$$V_{N,i} = \max(S_{T,i} - K, 0). \tag{39}$$

4. **Backward induction:** The option value is computed recursively from the final nodes back to the root using risk-neutral valuation:

$$V_{n,i} = e^{-r\Delta t} \left(qV_{n+1,i} + (1 - q)V_{n+1,i+1} \right). \tag{40}$$

5. **Initial price:** The value $V_{0,0}$ at the root of the tree represents the option price at $t = 0$.

6.3 Numerical experiment

In our implementation, we utilize $N = 500$ time steps. The resulting binomial price is compared against the analytical Black–Scholes price to verify convergence. We observe that as N increases, the binomial price approaches the analytical solution, even more than the FDM.

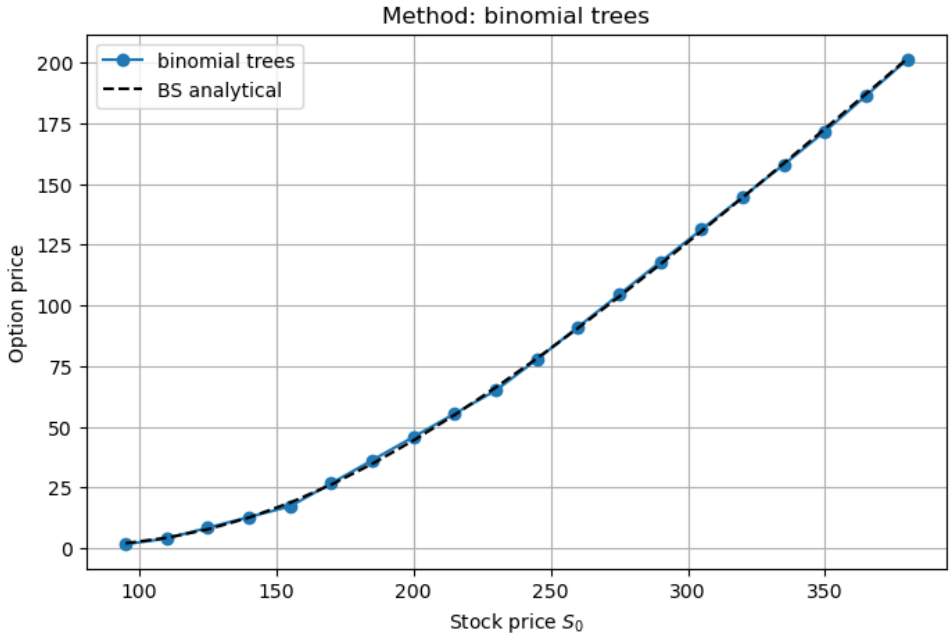

Figure 8: Binomial trees and BS